

GPS_GNSS_selection_no_solar

GNSS/GPS Positioning Options

Positioning Requirement

The animal tracker needs at least one position/activity datapoint per minute. Since solar energy is not allowed, the positioning subsystem must minimize both active energy and stored/transmitted data.

The main design question is whether "one datapoint per minute" requires a fresh GNSS fix every minute, or whether a low-energy GNSS snapshot can be post-processed by the base station.

Candidate Components

Option	Component / method	Type	Size and mass	Power / energy	Stored data per point
A	Quectel LC76G(PA)	Conventional GNSS module, external antenna	10.1 x 9.7 x 2.4 mm, 0.3 g; antenna not included	About 33 mW during acquisition/tracking; backup about 13 uA	20-40 B after the module computes latitude/longitude
B	Quectel LC86G series, especially PA/LA variants	Conventional GNSS module with integrated patch antenna	Integrated antenna family, about 16-18.4 mm square; LA approx. 8 g	PA tracking current about 11 mA at 3.3 V; other variants about 30-34 mA	20-40 B after the module computes latitude/longitude
C	SnapperGPS / Snapshot GNSS	Records raw GNSS signal snapshot, position solved later by base station/cloud	SnapperGPS PCB v2: 36.4 x 31.2 mm; battery-dependent mass	Less than 0.3 uAh per snapshot, about 0.004 J at 3.7 V	About 6 kB raw data per snapshot

Energy Comparison for 1 Position per Minute

Assume 1440 position samples per day.

Method	Energy per sample	Daily positioning energy	Comment
Conventional GNSS, optimistic 7 s active time at 33 mW	0.23 J	331 J/day	Requires excellent antenna and assisted/warm starts
Conventional GNSS, literature value from thermal tracker	0.621 J	894 J/day	Freiburg tracker achieved one fix every 1.1-1.5 h, not every minute
Snapshot GNSS	about 0.004 J	about 5.8 J/day	Much better for no-solar operation

Conclusion: conventional GNSS is suitable for low-frequency fixes, but not for a fresh fix every minute in a no-solar energy-harvesting design. Snapshot GNSS is the only positioning option in this list that makes minute-scale sampling energetically realistic.

Data Storage and Transmission

Conventional GNSS

The GNSS module outputs computed position data. A compact record can contain timestamp, latitude, longitude, fix quality, activity state, and battery voltage.

```
Record size:          about 20-40 B
1 day at 1/min:       28.8-57.6 kB
3 days at 1/min:      86.4-172.8 kB
```

This is compatible with local flash and later LoRa batch upload.

Snapshot GNSS

Snapshot GNSS stores raw signal data instead of latitude/longitude.

```
Snapshot size:        about 6 kB
1 day at 1/min:       8.64 MB
3 days at 1/min:      25.9 MB
7 days at 1/min:      60.5 MB
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A SnapperGPS-style board with 22,000 snapshot capacity stores about 15 days at one snapshot per minute.

This data volume is not suitable for LoRa. LoRa should only transmit status packets, alarms, or a small number of selected records. Snapshot GNSS would require a higher-data-rate local upload method, for example Wi-Fi, Wi-Fi HaLow, BLE at short range, or physical retrieval.

Sources

- Quectel LC76G series: <https://www.quectel.com/product/gnss-lc76g-series/>
- Quectel LC86G series: <https://www.quectel.com/product/gnss-lc86g/>
- Quectel GNSS product overview: <https://www.avnet.com/wcm/connect/9d7ace1a-3bff-4742-84c1-248fd9271190/quectel-gnss-module-product-overview.pdf>
- SnapperGPS PCB v2: <https://github.com/SnapperGPS/snappergps-pcb-2>
- SnapperGPS paper: <https://ojs.lib.uwo.ca/index.php/openhardware/article/view/17860/13495>
- Baeumker et al., thermal wildlife tracker: <https://doi.org/10.3390/en14196363>